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Work package E: Tracking ships

Deliverable E1

Defining products for AIS implementation

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Prepared by:

Anke Consten (CBS, The Netherlands)
Dennis Dahlmans (CBS, Netherlands)
Wai King Meijer-Cheung (CBS, Netherlands)
Sjoerd Schenau (CBS, Netherlands)
Mark van der Spoel (CBS, Netherlands)
Tessa de Wit (CBS, Netherlands)
Eleni Bisioti (Elstat Greece)
Christina Pierrakou (ELSTAT, Greece)
Michal Bis (GUS, Poland)

Work package Leader:

Anke Consten (CBS, The Netherlands)
acon@cbs.nl

telephone : +31 45 570 7335
mobile phone : +31 6 5248 7727

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Executive summary

Building on the developments of the completed work in the ESSnet big data programme I [1] on the use of AIS data (Automatic Identification System) for maritime statistics, the aim of this Work Package: tracking ships (WPE) is to develop functional production prototypes. Results of the WP in the first program showed the potential wealth of AIS data to improve current statistics and to generate new statistical products. Although some important elements of current maritime statistics such as type and quantity of goods loaded or unloaded at the port are not part of AIS, AIS still is useful to improve other aspects of statistics and provide new products. In addition, AIS data is also useful for statistics on inland waterways, emissions and can provide other new products. Concerns signalled were the need for suitable hardware and software tools for the exploration and exploitation of the huge amount of AIS data and the lack of documentation and guidance for using AIS.

This deliverable describes two tasks of this work package E: Tracking ships (WPE), i.e. data access and description of the product definition. The latter describes the results of the definition phase: which statistical products/prototypes to be delivered, and in which statistical processes to implement AIS data (Automatic Identification System).

Data access

AIS data is an internationally standardized, encoded data source available for national territories, the entire European territory, as well as at a worldwide level. This WP aims to implement AIS both in maritime statistics and in inland waterway statistics. European AIS data on maritime ships and national data on inland waterway ships has to be available for this. The European Maritime Safety Agency (EMSA) has been requested to share European AIS data from 2015, 2016, 2017 and 2018 on Maritime ships. For this project, we received AIS data covering 2016 and 2017. Dutch AIS data on inland waterway shipping for 2017 has been be provided by Rijkswaterstaat[1] (RWS). The latter will only be available at Statistics Netherlands.

The maritime AIS data from EMSA has to be accessible for all group members. Therefore, it will be accessible at Eurostat's Big Data Test Infrastructure (BDTI). For now, we have access to the BDTI from May to October 2019. For 2020, we will have to submit a new request for access to and use of the BDTI. This poses a risk to the project.

Product Definition

Aim of this task is to define the specific outcomes of this WP: the statistical products/prototypes (both new statistics and improved existing statistics) and in which statistical processes AIS is to be implemented. Four main products have been defined:

1. **Improve quality of statistics on port visits:** Use AIS data for the compilation and verification of existing Maritime Transport Statistics (MTS). It will be investigated whether the number of specific vessels calling at a port can be derived by AIS data.

[1] Rijkswaterstaat is responsible for the design, construction, management and maintenance of the main infrastructure facilities in the Netherlands. This includes the main road network, the main waterway network and water systems. https://www.rijkswaterstaat.nl/english

If this verification process is successful, it leads to a faster production tool for Greece and perhaps some other countries. It will also reduce the time and resources needed for processing data and increase the quality of statistics.

2. **Improve statistics on Inland Waterway transport:** Currently, statistics on Dutch Inland Waterways (IWW) are based on incomplete information. Information on certain ships, some parts of the Dutch inland waters and some goods carries is missing. AIS enables us to complete information on missing ships and estimate (based on type of terminal) type of goods ships might have carried.
3. **Improve statistics on air emissions and energy used**: Using AIS data we would like to investigate whether we can estimate the annual energy use and the air emissions from sea and coastal passenger and freight transport and Fishery of a) Dutch residents' vessels and b) vessels that operate under the Dutch flag. Currently there is no data available for the emissions outside of Dutch territory of Dutch residents vessels for deep sea shipping.
4. **Develop statistics on the behaviour of fishery ships:** The use AIS data will be investigated to study the behaviour of fishery ships. This should lead to experimental statistics in the domain of maritime fishing fleet, resulting in automated methods for detecting potential fishing behaviour. The resulting prototype will be replicated for the European Statistical System and other National Statistical Institutes (NSI's).

Besides these implementation pilots, further investigations will be performed to see whether or not it is possible to:

- Develop an algorithm to construct a reference frame of ports
  This algorithm replaces manual defining all ports. Manually defining all ports is a costly undertaking as there is such a high number of ports, and due to growth or shrinkage of ports, the location of ports can change.
- Improve the port to port distance matrix
  The port to port matrix is used to calculate tonne-kilometres (an important performance indicator for transport statistics). The matrix is now only partly based on a limited number of AIS travels. Taking into account more variables and travels would improve greatly improve the matrix.
- Use AIS as a fast indicator for international trade
  The number of ships visiting a port could be related to international trade. This could provide a "fast" indicator source that is closely related to economic development.

It is not foreseen that the outcomes of these investigations will directly be implemented in one of the statistical offices during this ESSnet, but outcomes should lead to valuable insights for the ESSnet as a whole.

This deliverable provides a basis for this WP by defining the statistical products/prototypes to be delivered and in which statistical processes to implement AIS data. In the next steps of this project we will build prototypes, resulting in experimental statistics for all four processes. The final aim of this WP is to fully implement AIS data into at least one statistical process. For each of the identified processes there will be at least:

- A description (process model) on the future process, including tools, infrastructure and conditions required (for example hardware requirements and skills needed).
- A report describing activities needed for developing experimental statistic into a solid process

To deliver the results planned, it is important to have an up and running Big Data Test Infrastructure (BDTI) including European AIS data and tools. To improve the quality of the port visits statistics it requested to have access to technical ship characteristics (IHS Maritime and Trade (Information Handling Services) or Marinfo -). This is also important to calculate fuel used and emissions. For this we see a risk in obtaining intermediate products from EMSA. In addition, having worldwide AIS data would be required to get optimal results.

1. Introduction

Building on the developments of the completed work in the ESSnet big data programme during 2016-2018 [1], the aim of this work package (WP) is to develop functional production prototypes. Results of the WP in the first program showed the potential wealth of AIS data to improve current statistics and to generate new statistical products. Although some important elements of current maritime statistics such as type and quantity of goods loaded or unloaded at the port are not part of AIS, AIS still is useful to improve other aspects of statistics and provide new products. In addition, AIS data is also useful for statistics on inland waterways, emissions and can provide other new products. Concerns signalled were the need for suitable hardware and software tools for the exploration and exploitation of the huge amount of AIS data and the lack of documentation and guidance for using AIS.

This included setting up procedures and developing technical solutions. It also includes promoting and supporting data collection, processing and analysis of (big) data from AIS (Automatic Identification System (AIS) for statistical production in the participating national statistical institutes (NSI's) related to statistics on maritime, inland waterways (IWW) and environmental statistics. Three National Statistical Institutes participate in WPE: the national statistical institutes of The Netherlands (Work package leader), Greece and Poland.

The output of the work should become available to the European Statistical System (ESS) and enable other NSI's, in the next implementation phase, to produce statistical output. Therefore, active involvement of subject matter experts, statisticians, enterprise architects and other stakeholders is already needed. Subject matter experts for the NSI's working in the relevant national statistical production units (e.g., maritime and IWW transport, emissions, fishing) will be informed in all phases of work. Informing relevant Eurostat Working groups and Task Forces (e.g., maritime and IWW transport) will be part of the work of this WP.

Along the production of the relevant methodologies, recommendations, specifications and statistical software, the production of experimental statistics demonstrating the capabilities to produce statistics is an objective of this WP.

Methodological, qualitative and technical results of this WP, including intermediate findings, will be used as inputs for WPF (on the definition of reference architecture to carry out big data production both at national and European level) and WPK (on general methodology and quality of big data in the statistical production process). When carrying out the tasks listed below, care will be taken that these results will be stored for later use, using the facilities described in WPA (on coordination and communication).

The current WP will result in four deliverables, of which this deliverable is the first one:

- E1 (May 2019): Defining products for AIS implementation (excluding Reference Methodological Framework and data models)

- E2 (March 2020): Outcomes of prototypes. Describing aspects such as implementation requirements at European and national level, quality framework, required metadata and a risk plan. The content of this report is based on the results of the prototypes (task 4).

- E3 (February 2020): Interim technical report. Describing the conditions for data, methodology and procedures to be used for producing statistics (development of functional prototypes, development toolkit and software), recommendations and guidelines for full-fledged implementation in the ESS.
- E4 (October 2020): Final detailed report (including Reference Methodological Framework and data models) describing the outcomes of this WP. The results will also include an assessment of sustainability over time.

See Annex 1 for a more detailed description of the work in this WP.

The aim of this first deliverable (E1) is to describe data access and the definition phase: which statistical products/prototypes to be delivered and in which statistical processes to implement AIS data. This is mostly based on a meeting In February 2019 where WP members from the different NSI's discussed the use of AIS in statistical processes [2].

In this deliverable two tasks of this WP are described. To begin with, task 2 of the project, data access is described in chapter 2. Chapter 3 describes the product to be delivered for this work package. Finally, in chapter 4 the overall conclusions of this deliverable are provided.

2. Data access

AIS data is an internationally standardized encoded data source available for a worldwide level. This WP aims to implement AIS in maritime statistics as well as in IWW statistics. Therefore, European AIS data on maritime ships as well as national data on IWW ships have to be available. Aim of Task 2 is to obtain these data sets. The AIS data source used in the ESSnet big data programme 2016-2018 did not have sufficient coverage, and data was not sufficiently up-to-date to use for implementation. Instead of using the same source, we now requested non-commercial European AIS data from the European Maritime Safety Agency (EMSA). We expect this data to have sufficient coverage to produce statistics. We requested data for maritime ships from 2015 until 2018. For now, we only have received AIS data covering 2016 and 2017. In case we need data from 2018, we will submit a new request.

Dutch AIS data on inland waterway shipping for 2017 has been provided by Rijkswaterstaat (RWS). This data will only be available at Statistics Netherlands.

Maritime data from EMSA will be used for the statistics on port visits and fishery fleet. Originally, we also planned to use the data for emission statistics. However, EMSA performs calculations on emissions for European countries. Therefore, we will investigate the usability of EMSA's methods instead of developing methods to calculate distances travelled and emissions ourselves [1]. If usable, we will request EMSA to provide us with intermediate products needed for our statistics. In all cases, we want to prevent redundant work. This only has consequences for the implementation of AIS in the environmental accounts: calculate fuel consumption and emissions.

Maritime AIS data from EMSA has to be accessible for all group members. Therefore, data will be accessible at Eurostat's pilot Big Data Test Infrastructure (BDTI) for cloud computing. As the infrastructure is part of a pilot that has limitations in terms of usage, we do not have access during the full running time of the project. For now, we have access from May to October and perhaps some time in the fall of 2019. For 2020, we have to submit a new request for access to and use of the BDTI. In addition, it is not possible to work on the data for more than one person at the same time. These factors pose a risk to this project, as working time is restricted

3. Products to be delivered

Aim of task 3 is to define the specific outcomes of this WP: the statistical products/prototypes (both new statistics and improved existing statistics) and in which statistical processes to implement AIS. If possible, prototypes will be tested by other group members work package. We will also describe how to involve subject matter experts, statisticians, enterprise architects and other stakeholders (Working Groups of inland waterways and maritime statistics and possible Task Forces) for other member states.

We have defined four main products. The first builds on work done in the previous ESSnet: improving the quality of statistics on port visits in Greece. The second product involves the implementation of AIS in the process of producing statistics on IWW transport in the Netherlands. As this process is very country specific, it will only be implemented for the Netherlands. The third product will improve the methods to calculate energy use and emission statistics for maritime and fishery ships owned by Dutch enterprises. The fourth product will be experimental statistics, series of statistics that are in the testing phase and not yet fully developed, on the behaviour of fishery ships.

Improve quality of port visits statistics

Definition of statistical products/prototypes

The primary aim of this prototype is to use AIS data for the compilation and verification of existing Maritime Transport Statistics (MTS) under the Directive 2009/42/EC [3]. From the previous ESSnet it seems possible to use AIS data to improve the quality and timeliness of statistics on port visits. Statistics on port visits, consist of a data set on ships visiting ports to load or unload cargo in the main European ports. The number of visits at port level are broken down into port, type (e.g., bulk or containership) and size (in gross tonnage, GT) of vessels that are loading or unloading cargo, embarking or disembarking passengers. This includes cruise passengers on cruise passenger excursions. Each member state has to deliver this so-called F2-table to Eurostat, See Table 1.

Data set F2: Vessel traffic in the main European ports, by port, type and size of vessels loading or unloading cargo, embarking or disembarking passengers (including cruise passengers on a cruise passenger excursion)

Periodicity of data transmission: annual

	Variables	Coding detail	Nomenclature
Dimensions	Data set	Two-character alphanumeric	F2
	Reference year	Four-character alphanumeric	(e.g. 1997)
	Reference quarter	One-character alphanumeric	(1, 2, 3, 4)
	Reporting port	Five-character alphanumeric	Selected EEA ports in the port list
	Direction	One-character alphanumeric	Inwards (1) only
	Type of vessel	Two-character alphanumeric	Type of ship, Annex VI
	Size of vessel GT	Two-character alphanumeric	Gross tonnage size classes, Annex VII

Data: Number of vessels.
Gross tonnage of vessels.'

Table 1: Structure of the data set F2

We want to further investigate whether the number vessels visiting a port can be derived by AIS data,

focusing on ship's characteristics. AIS data contains information on the type of vessel. However, the classification of 'Type of vessel' of AIS data is not as detailed as required by the Directive 2009/42/EC. For example, Eurostat distinguishes the ship categories "container", "dry bulk" and "liquid bulk" whch are all part of the broader category "cargo ship" in AIS. This is due to the primary use of AIS data being safety at sea, not production of statistics. Therefore, we plan to examine whether the analytical types of vessel can be derived from another data source (such as IHS Maritime and Trade, Information Handling Services) for completing this variable in the F2 data set. The same is true for the variable "Size of vessel gross tonnage". Our plan is illustrated in the following Figure 1.

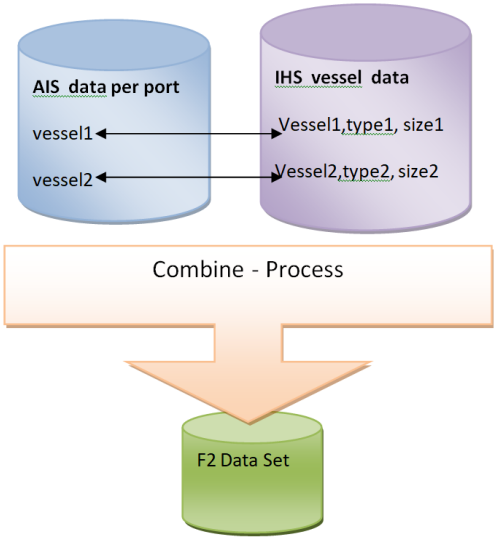

Figure 1: Combination of AIS and IHS Maritime and Trade data for the production of F2 data set

AIS will be used to validate and generate statistics for the Maritime Transport Statistics under Directive 2009/42/EC. AIS is relevant for both the national and European level. The result of this process is statistics on seagoing vessels that carry goods and/or passengers, with a gross tonnage of 100 and more. Each member state collects data on loading/unloading of goods and passengers by seagoing vessels docking in their territorial ports [4]. This F2-dataset is mandatory for "main ports". Smaller ports are allowed to report on a voluntary basis. At a national level, for each port selected, detailed maritime transport data is collected by the national competent authorities using a variety of data sources, such as port administration systems, questionnaires to ports, shipping companies and shipping agents [5].

Improvements by using AIS

If this verification process is successful, it can be included in the production as a faster tool for Greece, since this process is currently performed on a manual basis based on printed reports of port calls. It will also reduce the time and resources needed for processing data and coverage of statistics, that is more ports will be covered. For some ports it may even provide a better number of ship visits, as some port authorities work together and do not distinguish between ports. A ship that visit these ports will then be counted to have visited both ports.

Preconditions for this product

For the development of the F2-data set access to data set such as IHS Maritime and Trade or Marinfo is needed to add information on type and size of all vessels.

First steps

The aim is to develop a prototype for the verification of F2 data set for a main port of Greece (e.g. Piraeus) for a specific period (e.g. a quarter).

a) A reference frame of vessels from AIS data will be developed, to select maritime ships.
b) Enrich the reference frame with ship information from the national database containing type and size of vessels available in ELSTAT.
c) Calculate the F2 data set from AIS combined with the reference frame data and compare the results to data from ELSTAT's Maritime Transport Unit.

Usability

Since the F2 data set is mandatory for member states, and the data are collected at a European level, the prototype could be used by other NSI's in the ESS. In the future, using AIS data for the production of maritime statistics with common sources and methodologies at the ESS level will improve the accuracy, comparability and timeliness.

If this verification prototype proves to be reliable and faster than currently in place, coverage could be extended to other Greek ports. The usability of this prototype will also be tested from one of the other NSI's in this WP. All algorithms developed under WPE will be available at Github.

Involvement of stakeholders

Important stakeholders for this part of the project are:

a. ELSTAT Maritime Transport Unit - Provides data concerning type and size of vessels available in ELSTAT and active involvement on the development of the prototype
b. Port authorities of Piraeus – Assistance in Port functions and services and potential data suppliers
c. EMSA - AIS data supplier for the years 2016 and 2017.

We will contact these stakeholders and inform them on progress of this project.

Risks

The national database of ship characteristics currently available on ELSTAT's Maritime Transport Unit is only available for vessels under Greek flag, it is not expected to cover all the vessels visiting Greek ports. One solution could be to get access to a European data set such as IHS Maritime and Trade which contains information about type and size of all vessels worldwide. This will have financial implications.

AIS data might reveal more arrivals of vessels than the ones of MTS, because some vessels may not approach a port for commercial purposes of loading and/or unloading cargo or embarking and disembarking passengers as required by the regulations. In case this occurs frequently, extra information from port authorities will be needed.

Improve inland waterway statistics

Definition of statistical products/prototypes

For the Netherlands, statistics on Inland Waterways (IWW) is based on incomplete information. Information on certain ships, its goods and ships in some parts of the Dutch inland waters is missing. We aim to complete the largest part of this missing information using AIS.

Furthermore, we aim to develop several extensions for current statistical products, including traffic intensities on inland waterways. This helps us to better quantify transport of specific goods in specific regions. This in turn can help us to calculate travelled distances, which can be the basis for emission calculations for IWW. Also, compulsory table II2, could be created using AIS data. This table contains the number of ships travelling and the distances travelled by these ships. This includes both loaded and empty ships.

The current statistical process on IWW transport will be affected on both European and national level as we want to update the current Eurostat tables and StatLine tables. The main purpose of this product is to improve the current statistics on transport of goods on inland waterways.

Improvements by using AIS

The current data is provided by Rijkswaterstaat (RWS) and comprises of ships travelling across locks in inland waters. Coverage of this data source for IWW statistics is known to be incomplete for parts of the Dutch inland waterway network. As ships can pass through the rivers without passing a lock, ships can take routes that do not require any reporting to RWS. These routes will now be covered by AIS. Furthermore, it is not obligatory for certain ships to report goods transported to RWS, thus some information on goods transported is missing. Using information from other ships, we can characterize the type of terminal, which can provide useful input to estimate the type of goods ships have carried.

Preconditions for this product

In the past, research performed for RWS has been conducted on a single month of AIS data. The methodology and findings of this study need to become familiar for the current project team. In addition, a larger AIS data set covering two years is needed. Finally, a data and software infrastructure has to be developed at Statistics Netherlands.

First steps

First we want to validate methods Statistics Netherlands developed for an earlier project performed for RWS. In that project, the use of AIS was investigated for RWS's processes. Then, we want to investigate whether the methods developed work for a larger dataset. This will probably result in further improvements of the methods. The next step is to describe the current statistical process for IWW. This will result in the decision on where to implement AIS: as there are different possibilities on combining AIS to data from the original source (data from locks). The latter will remain the basic source of the statistics on IWW.

Usability

The case at hand is very specific for the process of generating Dutch IWW statistics. As other countries have different methods to generate statistics on IWW, it will not be possible to test our methods directly for other countries. An extra complication to this is the availability of AIS data on IWW, as we only have AIS data for the Dutch IWW network. However, we are planning to generate methods and modules for specific steps in the process. For instance, to decode AIS data, we will adapt the available Python AIS module. Also, for identifying goods a more generic method will be developed. These methods can be tested and possibly be applied by other NSIs.

Involvement of stakeholders

RWS has worked on implementing AIS in the original data source from locks, that will remain the basis for this statistic. In contrast to our current project, they have not used raw AIS data. Issues concerning completeness have not yet been addressed. To learn from each other, we will discuss both our findings

as well as their findings in a meeting. However, our case goes further than RWS' case on completing data on the number of ships passing a lock (see also *Risks*).

Eurostat will obtain updated data for the IWW statistic, in which the incompleteness problem is solved. When our plans are implemented, and a feasible prototype has been manufactured, we will present our work during a Working Group meeting at Eurostat. Possibly, other NSI's can apply some of the algorithms we developed.

Risks

The data for the Dutch inland waterway network is provided for 2017 by RWS on a Hard Disk Drive. This is sufficient for this prototype, but in the future a more sustainable data delivery system needs to be created. Future delivery plans need to be formalized.

There are some doubts about the hardware capacity to trim down the raw AIS data to a usable format, however alternative resources are available within short notice.

Currently there is a small risk of redundant work, because RWS is also working on implementing AIS into the data source. It is possible that they completely address the incompleteness issue, thus making this project unnecessary. However, we do not expect this to be done in the near future. In addition, our implementation also aims at completing information on missing goods, regionalization and traffic intensity. RWS does not focus on these products.

Improve statistics on air emissions and energy used

Definition of statistical products/prototypes

Using AIS data we want to investigate whether we can estimate the annual energy use and the air emissions from Sea and coastal passenger and freight transport (NACE H50.1 and NACE H50.2, in the code of the statistical classification of economic activities in the European Community) and Fishery (NACE A3.1) of a) Dutch residents' vessels and b) vessels that operate under the Dutch flag.

The method will follow the residents principle of the National Accounts (SNA) and Environmental Accounts (SEEA), and the estimations will be part of the NACE 64 industries and households allocation. Energy account and Air emission account are part of the Environmental Accounts (SEEA), which is a common framework to establish the link between the environment and the economy. For the energy account, the results will be in energy used by fuel type, in petajoules (PJ). For the air emission account the results will be for four pollutants: carbon dioxide(CO2), coarse particulate matter (PM10), mono-nitrogen oxides (NOx) and sulphur dioxide (SO2), in million kilogrammes.

The air Emission Account and Energy Account from the Environmental Accounts (SEEA) will be affected on national level, and also on European level when the dataset is aggregated by Eurostat into EU statistics.

These two accounts are compulsory Environmental Accounts statistics of Eurostat. Annually all EU countries have to submit in the form of questionnaires the national dataset of the years 1995 to year T-2, allocated to NACE groups and households.

Improvements by using AIS

Currently there is no data available on the emissions outside of Dutch territory of Dutch residents'/companies' vessels for deep sea shipping. Estimations for the air emission account are based on National Accounts data (fuel used in monetary value). For air emission statistics following the territory principle, AIS data is already used for estimations on Dutch territory/coastal emissions, however this data includes non-residents vessels. Data on fuel used by Dutch fishery ships is sometimes available (from Wageningen Economic research institute). However, this data is not annually available and continuity is unclear. As the method for fishery will most likely be not too divergent from the method for deep sea shipping, this industry is added to this project. For air emission statistics following the territory principle, AIS data is already used for estimations of Fishery on Dutch territory. These estimations are calculated by TNO (The Netherlands Organization for applied scientific research) & MARIN (Maritime Research Institute Netherlands).

Preconditions for this product

Data sources have to be available. For this process in principle, this has to be data that covers worldwide AIS signals.

We have to see whether EMSA can provide the energy used of Dutch maritime and fishery ships (an intermediate product), and emissions of these ships.

First steps

As EMSA calculates emissions from AIS, we will investigate the possibility of obtaining data on energy used from EMSA. We will also try to get access to worldwide data. We will also investigate which ships are linked to Dutch economy and corresponding definitions. In addition, we will investigate which definition of Dutch owned ships on Ministries' list reflects economical value best. Also we will investigate if nationality of fishery ships can be derived from the ship's flag. The next step then will be to see how to share the list with Dutch owned maritime ships with EMSA to get the distances travelled and the emissions.

Usability

The method and algorithm will be openly available. Not all source files of the Netherlands (such as residents' vessels with company information) can be shared due to Dutch government privacy rules. However, the aggregated results and analyses will be published as reference for their statistical offices.

Involvement of stakeholders

For this specific part of the project we identified the following main stakeholders:

- EMSA – Provider European AIS data on Maritime and Fishery ships. Provides data and assistance on the implementation of the method. EMSA is in the opportunity to attend the meetings of WPE and will receive all deliverables.
- EUROSTAT – User of energy account and air emission account. Eurostat will be continuously updated on the progress of this project.
- Dutch Ministry of Infrastructure and Water Management – User of energy account and air emission account. The department will be updated on the progress of this project.

Risks

We are not sure if we can obtain intermediate products from EMSA. This poses the first risk, as this would result in extra work on our side. Also, we doubt if we can calculate emissions, and we would have to lower the ambition and use travelled distance as a proxy of emissions instead.

Second, we still do not have access to worldwide AIS data. When only European AIS data is available (from EMSA) a big part of the energy use and air emissions will most likely be left out of the estimation. This will be insufficient for what is needed for the compilation of the air emission accounts and the energy accounts.

Identifying the vessels that are linked to Dutch residents shipping and fishery companies is another risk. There is an annual source available where all vessels linked to the Netherlands, by either flag type or ownership types are collected. The next step is to identify which of these vessels are producing economical value to the Dutch economy. These ships will be tagged in the AIS data. However tracing these particular vessels can be very difficult. There is annual financial and production data from residents' companies from the Production Statistics, including information such as annual turnover, operational costs etcetera. But no information is provided as to which vessels specifically achieve these results. Therefore tagging the relevant vessels will be one of the biggest challenges in this project, next to developing an algorithm to calculate energy used and air emissions from AIS data.

A fourth risk is that, developing a method to calculate fuel used per ship is very technical. This is part of the technical calculation method, where technical information is needed such as engine size, build year, route area (regulation) etcetera. We don't know if we have sufficient technical knowledge to developed this method or if we can apply already developed methods such as AIS methods from TNO.

Finally, for the calculation from fuel used into emission, the emission factors are needed. These technical factors are calculated by technical research institutes. The availability of these emission factors is essential for the calculation.

Develop statistics on fishing fleet

Definition of statistical products/prototypes

Goal 14 of the Sustainable Development Goals (SDGs) focuses on conserving and sustainable use of the oceans, seas and marine resources [5]. We want to investigate the use of AIS data to study the behaviour of fishing fleet. This should lead to experimental statistics in the domain of maritime fishing fleet. We will develop a functional prototype including procedures and software, to process and analyse maritime fishing fleet active for the coast of Poland. This should result in automated methods for detecting potential fishing behaviour. The prototype developed could be replicated for ESS and other NSIs.

Examples of indicators that we plan to develop:

- Number of active fishing fleets for a specific port and time period;

- Number of fishing fleets in port(s);

- Number of expected entries of fishing fleets to port(s);

- Average (max, min) time of activity of fishing fleets depending on the season and/or weather conditions;

- Average (max, min) distances travelled per fishing vessel depending on the season and/or weather conditions;

- Average (max, min) speed/draught per fishing fleet for a specific fishing areas and time period

traffic/intensity of fishing fleet in a particular area;

- Traffic/intensity of fishing fleet in a particular area.

Improvements by using AIS

Behaviour of fishing fleet indications will constitute a completely new product.

Preconditions for this product

We only need AIS data to develop statistics on fishing fleets. We are aware of the existence of other sources related to the activity of the fishing fleets, but focusing on the project's requirement for universality and replicability by other NSIs, the need to gain access to other sources could be a barrier to the adaptation of our solution by users.

First steps

We will start by defining areas of interest: fishing ports and fishing areas. Also, a reference frame of fishing fleet has to be generated. Then, algorithms to calculate indicators will be developed.

Parallel to this, we will contact the Fishery department of Eurostat to discuss our plans.

Usability

This project was selected because of the relevance to SDG's. Since most fishing vessels (even below 15 meters) use the AIS system for safety reasons, we propose to connect the AIS data with those from the fishing fleet register. As a result, we will be able to measure both the traffic and activity of Polish fishing fleet within the range of the AIS in the real time. The data collected and processed in that way would be a useful comparative tool, e.g. for the European Commission and its subsidiaries while satisfying marine environment requirements as listed mainly in: Commission Implementing Regulation (EU) 2017/218 of Feb 6 2017 on the Union fishing fleet register, the Common Fisheries Policy and the UN Goal 14 (Conserve and sustainably use the oceans, seas and marine resources for sustainable development) etc.

Combined with a source that is universal, the developed algorithms and methodology will be fully replicable for other statistical offices in the ESSnet. The algorithms developed will be tested within this project for the Netherlands and Greece.

Involvement of stakeholders

Stakeholders are the Ministry of Maritime Economy and Inland Navigation, Maritime Offices, Center for Fisheries Monitoring, National Marine Fisheries Research Institute, Maritime University of Szczecin, Statistics Poland. Statistics Poland, as well as the European Commission (Directorate for Maritime Affairs and Fisheries), Eurostat (Directorate for Sectoral and Regional Statistics). Our project will enable the stakeholders to verify the methodologies, their use and cross-check their statistics.

During the project implementation, the demand for information in this area for the different stakeholders will be investigated. Working meetings with Maritime University of Szczecin will be planned to revise the chosen methodology if needed.

Risks

The risk for the chosen process may be the fact that some of the fishing fleet (under 15 meters long) may not use an AIS transmitter. In that case, the results will not represent a complete picture of the fishing fleet phenomenon but only for units over 15 meters long. The fishing fleet register will be used

to study the number of fishery ships not using AIS. However, our preliminary tests show that most of those vessels use AIS anyway, mainly for security reasons.

4. Overall conclusion

This deliverable aims to describe the results of WPE's definition phase: which statistical products/prototypes to be delivered, and in which statistical processes to implement AIS data.

We conclude that AIS data will be implemented in the current statistical processes of Dutch Inland Waterways (national AIS data) and the current processes of air emissions and energy used by (maritime) ships owned by enterprises from a specific country. European AIS data is also used for compiling the maritime F2 table to validate and complete the port visits. Last but not least, experimental statistics on fishing behaviour will be developed based on European AIS data. All statistical processes developed within one NSI will be tested by another participant in this WPE (if possible), resulting a generic approach in the end for the ESS.

Besides these implementation pilots, there will be some further investigation to see whether or not it is possible to:

- develop an algorithm that can construct a reference frame of ports. A next step could be to construct a reference frame of terminals. Now ports and terminals have to be defined manually, which is not feasible for countries with many ports.
- Improve the port to port distance matrix by investigating more journeys and contextual factors (e.g., weather conditions). It could also result in using actual AIS data per journey instead of the average distance matrix.
- Use AIS as a fast indicator of international trade, where international comparisons can be made quickly. National offices are really interested in fast indicators of economic development.

It is not foreseen that the outcomes of these investigations will directly be implemented in one of the statistical offices during this ESSnet, but they will provide very valuable insights for the ESSnet as a whole.

This deliverable constitutes the basis for this WPE. Fully functional prototypes of all four identified statistical production processes will be developed. In addition, experimental statistics will be produced for all four prototypes by visualising the first results. The final aim of this WP is to implement AIS data into (future) statistical processes. For each of the identified processes there will be at least:

- A description (process model) on the future process describing the processes including tools, infrastructure and conditions (for example hardware requirements and skills needed.
- A report describing the activities needed to develop an experimental statistic into a working process

At least one of the four processes will be fully implemented, including: production of tools and software (source code) that can be applied by other NSI's, methodologies, developer's and user's handbook and a procedure for testing and maintenance.

For a more detailed description on the exact activities within this WPE see Annex 1.

Prerequisites for delivering the planned output in this project are an up and running Big Data Test Infrastructure (BDTI) including the tools needed to process, analyse and visualize the data. Secondly it is important to have the European AIS data available to all participants of WPE. Third, we see a risk in

depending on intermediate data from EMSA on emissions and fuel usage per ship, which we might not be able to obtain.

To improve the quality of the port visits statistics it is important to have access to a database with technical ship characteristics (such as IHS Maritime and Trade or Marinfo). This is also important to calculate fuel used and emissions. It For the latter, to get optimal results worldwide AIS data would be required.

5. References

1. Consolidated report on project results including a cost-benefit analysis of using AIS-data for official statistics, May 2018

2. Milestone EM1 Report the first internal WP-meeting, April 2019

3. Working Group on Maritime Transport Statistics 23-24 May 2016 https://circabc.europa.eu/webdav/CircaBC/ESTAT/MOTST/Library/Maritime%20Transport%20Statistics/Meeting%20of%2023-24%20May%202016/Final%20minutes.pdf

4. Modal share of freight transport to and from EU ports, (IP/B/TRAN/IC/2014-065), PE 540.350, 2015 (EN) http://www.europarl.europa.eu/RegData/etudes/STUD/2015/540350/IPOL_STU(2015)540350_EN.pdf

5. https://link.springer.com/chapter/10.1007/978-3-319-73521-4_7

6. http://ieeexplore.ieee.org/abstract/document/7840774/

Annex 1: scope of WPE in more detail

Work package number	**WPE**	Start Month: End Month:		**M1** **M24**
Title	**Tracking Ships**			
Partner (leader in bold) **Person-days**	**NL** **298**	EL 225	PL 170	

Objectives

Taking into account the developments of the successfully concluded work in the ESSnet big data programme during 2016-2018, the aim of this WP is to develop functional production prototypes including setting up procedures and developing technical solutions, to promote and support the collection, processing and analysis of (big) data from AIS (Automatic Identification System) for statistical production in the participating NSIs related to statistics on maritime, inland waterways and environmental statistics.

The output of the work should become available to the ESS and adequately enable other NSIs, in the next implementation phase, to put in place the necessary system(s) and deploy or adapt the proposed solution(s) in order to produce statistical output. From the practical point of view, the output should in addition develop key artefact like target architecture definition and requirements and guidance for the next implementation and deployment phase, hence producing specifications for full-fledged implementation at the ESS level. Therefore, the active involvement of subject matter experts, statisticians, enterprise architects and other stakeholders (for example Working Groups of Inland waterways and maritime statistics, Transport and Environment Reporting Mechanism (TERM) and Task Forces) is important. Subject matter experts for the NSIs working in the relevant national statistical production units will be involved in all phases of work. Participation to relevant Eurostat Working groups and Task Forces will be part of the work of this WP. This includes presentation of findings and work to the relevant working groups at an early stage.

Along the production of the relevant methodologies, recommendations, specifications and statistical software, the production of experimental statistics demonstrating the capabilities to produce statistics is a prominent objective.

Methodological, quality and technical results of this WP, including intermediate findings, will be used as inputs for WPF and WPK. When carrying out the tasks listed below, care will be taken that these results will be stored for later use, by using the facilities described at WPA.

Description of work

With respect to the development and implementation of functional production prototypes according to the work carried out during 2016-2018 by the preceding ESSnet, this WP aims at completing the following activities that would lead to the implementation and integration of AIS data within the national statistical production process(es) in the NSIs:

Task 1 - Interaction with WPF (NL)

Aim of this task is coordinating the interaction with WPF. The aim of WPF is the definition of reference architectures necessary to carry out big data production both at national and European levels. This task includes (i) designing and adopting application and information architectures (WPF Task 1) and (ii) providing input to WPF Task 2.1 on solution architectures.

Task 2 - Data access (NL)

AIS are internationally standardized encoded data available for national territories and the entire European territory, as well as at global level. The used European data source in ESSnet big data programme 2016-2018 isn't recent enough to use for implementation.

This WP aims to implement AIS in maritime statistics as well as in inland waterway statistics. That is why it is important to have European AIS data on maritime ships available as well as national data on Inland waterway ships. The European Maritime Safety Agency (EMSA) was requested for European AIS data from 2017 and 2018 on Maritime ships. Dutch AIS data on inland ships will be provided by Rijkswaterstaat for implementing AIS in the statistical process of inland waterways.

This task aims to:

- Collect the data on maritime ships at a European level and collect the national data on Inland waterway ships for the Netherlands
- Set a sustainable long term collaboration frame with EMSA for future use at European level. This will require clear roles related to data, pre-processing, processing and production of statistics. Preparation of this future use of European AIS data from EMSA is also part of this task.

For this project it is essential to get at least access to European AIS data (from EMSA). At this point it seems that EMSA could provide the data needed for this project, but there is no final agreement yet. Costs for buying European AIS data from a commercial organisation, in the case EMSA cannot provide the data, are not budgeted yet.

Task 3 - Definition (EL, NL, PL)

Aim of this task is to define the specific outcome of this WP. What would be the delivered statistical product (both new statistics and enhanced existing statistics) at the end of this ESSnet? In what statistical processes AIS would be implemented? How would subject matter experts, statisticians, enterprise architects and other stakeholders (Working Groups of inland waterways and maritime statistics and Task Forces) be involved?

Key elements of this task are:

- Definition of statistical products/prototypes. Including a plan for possibilities for testing these prototypes in other countries.
- Identification of statistical production processes and capabilities that may be affected at national level

- Guideline for involving subject matter experts, statisticians, enterprise architects and other stakeholders. This includes involving subject matter experts for the NSIs working in the relevant national statistical production units in all phases of work as well as participation to relevant Eurostat Working groups and Task Forces. EMSA will follow the progress during this project. EMSA is also available for (methodological) advice.
- Description of an adequate Reference Methodological Framework for processing AIS data in statistical processes. Setting a sustainable long term collaboration frame with EMSA for future use of AIS data at European level will be included in building the necessary reference architecture using European data.

- Development of data models (output in XML or other standard), data verification and data exchange protocols for all the selected statistical processes for implementation

Task 4 - Prototypes (EL, NL, PL)

The aim of this task is the development of fully functional prototypes of the identified statistical production processes in task 3. Including a description on the adaptation and consolidation of the methodology, the procedures and the tools for collecting. Output would include software as well. This task also includes a definition of:

- the implementation requirements (report) of prototypes in the relevant statistical production processes at European and national level (all)
- a quality management framework (report) to ensure quality of outputs and process, leading to complementary statistics (indicators) (EL, NL)
- required metadata throughout the process (report, metadata schema) (all)
- a risk plan with mitigation scenarios (all)

Testing of the prototypes (software, architecture, technical requirements, etc.) will take place (if possible) in (at least) the participating countries of this WP. This testing phase would then allow corrections or development of alternatives to be adapted to national specificities of (at least) the partners in this WP.

Besides this, there will be taken into account a couple of recommendations resulting from the previous ESSnet on big data. These two topics will be further investigated in this ESSnet:

- Combining AIS with online information and/or information on IHS Maritime and Trade to provide more reliable information on the type and capacity of ships (NL, EL).
- Developing an algorithm for signalling anomalies in the movements of ships (e.g. different traffic intensities or travelling speeds) to get information on disruptions or (changed) regulations in ports and fairways (NL, PL).

The outcomes of the two above mentioned investigations will be really helpful for further implementation of AIS in official statistics in this ESSnet.

The three topics beneath will be further investigated in this WP as well:

- Developing an algorithm that can construct a reference frame of ports. The next step would be to construct a reference frame of terminals. Combining this with type of visiting ship could assist with ship visits where type of ships or goods is missing. Now ports and terminals have to be defined manually, this is not feasible for countries with many ports (NL).
- Improving the port to port distance matrix by Investigating more journeys and factors. Optimizing algorithms could also result in using actual AIS journey data instead of the average distance matrix. Optimizing algorithms could also result in using actual AIS journey data instead of the average distance matrix (NL).
- Using AIS as a fast indicator of international trade, where international comparisons can be made quickly. National offices are really interested in fast indicators of economic development (NL)

On some of these topics there are already some first findings available, resulting from the ESSnet big data programme 2016-2018. It is not foreseen that the outcomes of these investigations will directly be implemented in one of the statistical offices during this ESSnet, but they will provide very valuable insights for the ESSnet as a whole.

The results of all investigations mentioned will be produced as experimental statistics on Eurostat's dedicated section (task 4).

Task 5 - Experimental statistics (EL, NL, PL)

The aim of this task is to produce experimental statistics by visualising the first results of the outcomes on the prototypes in task 4. The experimental statistics will be published most likely in the form of visualisations and they will be hosted by Eurostat's dedicated section and on a national level.

Task 6 - Implementation (EL, NL, PL)

This task involves (the preparation of) implementing AIS in statistical processes. After finishing this task there will be at least:

- A description (process model) on the future process of all selected processes in task 2, describing the processes including tools, infrastructure and conditions (for example skills needed, hardware requirements).
- A report describing the activities needed for coming from the experimental statistic to the future process, for all selected processes in task 2. What needs to be done before the future process is ready for use and regular output can be published?
- One statistical process where AIS is fully implemented, including:

1. Production of tools and software (source code) that can be applied by other NSI's

2. Description of used methodologies
3. A developer's and user's handbook
4. A procedure for testing and maintenance.

Besides the above mentioned deliverables, the following more general actions will be performed in this task:

- Address issues related to sustainability of data sources, data use by NSIs, as well as data sharing between NSIs (apply risk management principles).
- Develop and evaluate scenarios for data governance and data management in the wider aspect of data sharing and collaboration processes management.

Task 7 - Future perspectives (EL, NL, PL)

Aim of this task is to summarise the project results in a final detailed report with the outcomes of this WP. Guidelines and documented material for full-fledged implementation in the ESS including relevant software. Setting a sustainable long term collaboration frame with EMSA for future use at European level will require clear roles related to data, pre-processing, processing and production of statistics. Preparation of this future use of European AIS data from EMSA will be will be part of this final report. The results will also include an assessment of sustainability over time.

Deliverables

E1 (M7)

- Report on the results of the definition process, concerning the Reference Methodological Framework and data models (results of task 3).

E2 (M12)

- Report on implementation requirements of relevant statistical production processes at European and national level, quality framework, required metadata and a risk plan (results of task 4).

E3 (M16)

- Interim technical report concerning the conditions for using the data, the methodology and the procedures to be used for producing statistics (development of functional prototypes, development toolkit and software), recommendations and guidelines for full-fledged implementation in the ESS

E4 (M24)

- Final detailed report with the outcomes of this WP. Guidelines and documented material for full-fledged implementation in the ESS including relevant software, description of used methodologies, developer's and user's handbook, procedure for testing and maintenance and description of skills needed. The results should also include an assessment of sustainability over time.

Milestones

EM1 (M9)

- Report on the WP meeting mid 2019

EM2 (M20)

- Report on the WP meeting mid 2020